

The Reactor Lineage

IndustrialCraft² → Reactor Incremental → Reactor Knockoff → Reactor Revived — a component reference and a family tree.

Where this game comes from

Reactor Revived is a clean-room rewrite of **Reactor Knockoff** (cwmonkey, 2013), which is a browser clone of **Reactor Incremental** (Cael, Kongregate), which is in turn an idle-game treatment of the nuclear reactor from **IndustrialCraft²**, the Minecraft mod. Four generations, and the physics survived all of them: you place fuel in a grid, adjacent fuel makes more power but far more heat, and everything else on the grid exists to move that heat somewhere it will not kill you.

IC² ships two reactor implementations, **Legacy** (the original, EU-only) and **Experimental** (the rewrite, which added fluid cooling and MOX fuel). The component set below is the Experimental one, which is what the descendants copied.

The law every generation inherited

IC²'s fuel rods produce **5 EU/t × n** and **2n(n+1)** heat per second, where *n* is the rod plus its adjacent rods. Power is **linear** in neighbours; heat is **quadratic**. That single asymmetry is the whole game — packing fuel together is always tempting and always punished — and every descendant kept it:

Game	Power from a cell	Heat from a cell	Shape
IndustrialCraft ²	5 × n	2n(n + 1)	linear / quadratic
Reactor Knockoff	base × (m + p)	base × (m + p) ² ÷ count	linear / quadratic
Reactor Revived	base × (m + p)	base × (m + p) ² ÷ count	linear / quadratic

n = this rod plus adjacent rods. *m* = the pack's cell multiplier (1, 4, 12), *p* = pulses from neighbouring cells, *count* = cells in the pack. Knockoff simplified IC²'s 2n(n+1) to a clean square, which is the same curve with a gentler constant.

Fuel: the numbers line up almost exactly

Pack	IC ² EU/t	IC ² heat/s	IC ² ratio	Revived power ×	Revived heat ×
Single rod	5	4	1 : 1	1	1
Dual rod	20	24	4 : 6	4	8
Quad rod	60	96	12 : 24	12	36

Power multipliers are **identical** across the two games — 1, 4, 12. Heat diverges: IC² runs 1, 6, 24 where the browser games run 1, 8, 36, because the square is steeper than 2n(n+1). Packing fuel is punished harder here than in Minecraft.

IndustrialCraft² reactor components

The grandfather's part list, with its published figures.

Fuel rods

Component	Output	Heat/s	Lifetime	Notes
Fuel Rod (Uranium)	5 EU/t	4	20,000 s	2,000,000 EU over a full cycle
Dual Fuel Rod	20 EU/t	24	20,000 s	Two rods in one slot
Quad Fuel Rod	60 EU/t	96	20,000 s	Four rods in one slot
Fuel Rod (MOX)	varies	varies	20,000 s	Output rises as the reactor gets hotter

Heat vents — destroy heat

Component	Cools itself	Pulls from reactor	Cools neighbours	Max heat
Heat Vent	6/s	—	—	1,000
Advanced Heat Vent	12/s	—	—	1,000
Reactor Heat Vent	5/s	5/s	—	1,000
Overclocked Heat Vent	20/s	36/s	—	1,000
Component Heat Vent	—	—	4/s each	—

Heat exchangers — move heat around

Component	With neighbours	With the reactor	Max heat
Heat Exchanger	12/s	4/s	2,500
Advanced Heat Exchanger	24/s	8/s	10,000
Reactor Heat Exchanger	—	72/s	5,000
Component Heat Exchanger	36/s	—	5,000

Storage, reflection and hull

Component	Figure	What it does
10k Coolant Cell	10,000 heat	Absorbs heat; cannot shed it on its own
30k Coolant Cell	30,000 heat	As above
60k Coolant Cell	60,000 heat	As above
RSH-Condensator	20,000 heat	Single use; recharged with redstone
LZH-Condensator	100,000 heat	Single use; recharged with lapis or redstone
Neutron Reflector	20,000 pulses	Boosts an adjacent rod without adding heat
Thick Neutron Reflector	120,000 pulses	As above, far more durable
Reactor Plating	+1,000 hull heat	Also reduces explosion radius by 5%
Containment Reactor Plating	+500 hull heat	Reduces explosion radius by 10%
Heat-Capacity Reactor Plating	+1,700 hull heat	Reduces explosion radius by 1%

A bare reactor hull holds **10,000** heat before it starts taking damage. Figures are from the official IndustrialCraft² wiki; the wiki flags its own fuel-rod page as outdated, so treat the rod mechanics as indicative of the Legacy behaviour the browser games were cloned from rather than of current IC².

What became what

Every IC² component, traced to the part it turned into in this game.

Component by component

IndustrialCraft ²	Reactor Revived	Tiers	How it changed
Fuel Rod (Uranium) ×1 / ×2 / ×4	Uranium Cell, Uranium×2, Uranium×4	3 packs	Kept whole. Six more fuel types were added above it: plutonium, thorium, seaborgium, dolorium, nefastium, protium.
Fuel Rod (MOX)	Protium Cell	3 packs	MOX rewards a hot reactor; protium instead gets permanently stronger each time one is spent.
Neutron Reflector / Thick	Neutron Reflector	6	Became a percentage power bonus to adjacent cells, with a finite pulse life, and a tier-6 variant that also adds heat.
Heat Vent (all five kinds)	Heat Vent	6	The five IC ² vent variants collapsed into one part with six tiers. The tier-6 Extreme Vent burns reactor power to do its cooling.
Heat Exchanger (all four kinds)	Heat Exchanger	6	Same collapse. Ours balances heat by <i>percentage full</i> across itself and its neighbours rather than by fixed rates.
Reactor Heat Exchanger	Heat Inlet / Heat Outlet	6 each	Split in two, and made directional: the inlet pulls heat out of neighbours into the reactor, the outlet pushes it the other way.
10k / 30k / 60k Coolant Cell	Coolant Cell	6	One part, six tiers. Still pure storage that cannot cool itself. The tier-6 Thermionic cell converts half of what it absorbs into power.
Reactor Plating (three kinds)	Reactor Plating	6	Kept as hull heat capacity. The explosion-radius mechanic was dropped — there is no world to damage.
RSH / LZH-Condensator	—	—	Dropped. Both are consumables you recharge by hand, which does not fit an idle game.
—	Capacitor	6	New. Raises the reactor's maximum <i>power</i> rather than its heat, which IC ² had no need for.
—	Particle Accelerator	6	New. Converts heat into Exotic Particles, the prestige currency. Explodes into an instant meltdown if it overheats.

What the browser games added

IC²'s reactor is a puzzle you solve once and then leave running. The idle-game descendants bolted on a progression: money and an auto-sell rate, a shop of ~60 upgrades, a reactor that grows from a fixed grid via expansion upgrades, and a prestige loop — bank Exotic Particles, wipe the board, keep the research. None of that exists in Minecraft, where the reactor is a machine in a world rather than the world itself.

Going the other way, plenty was dropped: fluid cooling and the Reactor Pressure Vessel, the crafting tree behind every component, depleted-rod reprocessing, and the fact that a real IC² meltdown takes a crater out of your base.

Sources — IC² figures: wiki.industrial-craft.net. Reactor Knockoff figures: read off the running game at cwmonkey.github.io/reactor-knockoff and its source. Reactor Revived figures: www.js/parts.js in this repository. Where a number is disputed between sources, the IC² wiki's value is the one printed here.